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Weight-space gradient descent induces approximate activity descent, making learning rules testable

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Abstract

A central question in neuroscience is whether biological neural networks optimise a loss function by adjusting synaptic weights – and if so, by what rule. Candidate rules are difficult to test because synaptic weights cannot be measured at scale in behaving animals, whereas neural *activities* can be recorded across learning. This raises the question: if a network performs gradient descent on its weights, what should we see in its activities? Here we show that weight-space gradient descent induces kernel-filtered activity descent; when the kernel is approximately diagonal, each neuron follows its own loss gradient. We validate this prediction in ANNs of moderate width and depth, showing that indeed changes in neural activities are correlated with their loss gradients. The same relationship can and should be tested in biological neural networks with existing experimental techniques.

Keywords: neural activity, activations, gradient descent, learning rules, neuroscience

Weight-space descent and observable activity. Neuroscience has long sought to understand learning rules in biological neural networks, and the success of ANNs trained by gradient descent has renewed interest in finding commensurate rules in the brain (Richards et al., 2019). A key challenge, however, is that synaptic weights are not directly observable in vivo; modern neuroscience experiments measure neural activity instead. Richards and Kording (2023) proposed validating gradient-based learning rules by observing how neural activity changes during learning, but implicitly assumed a link between gradient descent on synaptic weights and the resulting activity changes. We formalize that link here, showing that weight-space gradient descent induces kernel-filtered descent in neural activations.

Setup and derivation. Consider a feedforward network with layers $\ell = 1, \dots, L$, weights $\mathbf{W}$, and neural activities $A_i^{(\ell)}(\mathbf{W})$, trained to minimise a loss $\mathcal{L}$. Let $\mathcal{N}_\ell := \{1, \dots, n_\ell\}$ index the neurons in layer ℓ , let $\mathcal{P}_r$ index the scalar weights in layer r , and define the upstream parameter-index set $\mathcal{P}_{\leq \ell} := \bigcup_{r=1}^{\ell} \mathcal{P}_r$. Thus $i, j, k \in \mathcal{N}_\ell$ index neurons in a fixed layer, whereas $\alpha \in \mathcal{P}_{\leq \ell}$ indexes a scalar upstream weight W_α . We write $\mathbf{W}_{\leq \ell} := (W_\alpha)_{\alpha \in \mathcal{P}_{\leq \ell}}$ for the corresponding block of weights.

In gradient descent, every weight is updated simultaneously:

$$\Delta W_\alpha = -\eta \frac{\partial \mathcal{L}}{\partial W_\alpha}, \quad \forall \alpha \in \mathcal{P}_{\leq L}. \quad (1)$$

What is the induced change in the activity of neuron i in layer ℓ ? Because $A_i^{(\ell)}$ depends only on $\mathbf{W}_{\leq \ell}$, if the activity map is twice differentiable with bounded second derivatives along the update segment, Taylor expansion around the pre-update weights $\mathbf{W}_t$ gives

$$\Delta A_i^{(\ell)} := A_i^{(\ell)}(\mathbf{W}_t + \Delta \mathbf{W}) - A_i^{(\ell)}(\mathbf{W}_t) = \nabla_{\mathbf{W}_{\leq \ell}} A_i^{(\ell)}(\mathbf{W}_t)^\top \Delta \mathbf{W}_{\leq \ell} + O(\|\Delta \mathbf{W}_{\leq \ell}\|^2).$$

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Since $\Delta \mathbf{W} = O(\eta)$, substituting $\Delta W_\alpha = -\eta \partial \mathcal{L} / \partial W_\alpha$ gives

$$\Delta A_i^{(\ell)} \approx -\eta \sum_{\alpha \in \mathcal{P}_{\leq \ell}} \frac{\partial A_i^{(\ell)}}{\partial W_\alpha} \frac{\partial \mathcal{L}}{\partial W_\alpha}, \quad (2)$$

where all derivatives are evaluated at $\mathbf{W}_t$ and the omitted remainder is $O(\eta^2)$. For piecewise-linear activations such as ReLU, this statement applies when the step is small enough that no unit changes activation status, so the network remains within a single smooth piecewise-polynomial region of weight space. In a sequential feedforward network (one without skip or residual connections), the effect of any weight indexed by $\mathcal{P}_{\leq \ell}$ on the loss passes entirely through the activities of layer ℓ . This allows us to use the chain rule to write the effect on the loss, $\partial \mathcal{L} / \partial W_\alpha$, in terms of the activities at layer ℓ :

$$\Delta A_i^{(\ell)} \approx -\eta \sum_{\alpha \in \mathcal{P}_{\leq \ell}} \frac{\partial A_i^{(\ell)}}{\partial W_\alpha} \left[\sum_{k \in \mathcal{N}_\ell} (\partial \mathcal{L} / \partial A_k^{(\ell)}) (\partial A_k^{(\ell)} / \partial W_\alpha) \right], \quad (3)$$

Exchanging the sums yields

$$\Delta A_i^{(\ell)} \approx -\eta \sum_{k \in \mathcal{N}_\ell} \Phi_{ik}^{(\ell)} \frac{\partial \mathcal{L}}{\partial A_k^{(\ell)}}, \quad (4)$$

where

$$\Phi_{ik}^{(\ell)} := \sum_{\alpha \in \mathcal{P}_{\leq \ell}} \frac{\partial A_i^{(\ell)}}{\partial W_\alpha} \frac{\partial A_k^{(\ell)}}{\partial W_\alpha} = \nabla_{\mathbf{W}_{\leq \ell}} A_i^{(\ell)} \cdot \nabla_{\mathbf{W}_{\leq \ell}} A_k^{(\ell)} \quad (5)$$

is a Gram matrix defined on the activities within each layer, and $\partial \mathcal{L} / \partial A_k^{(\ell)}$ is the full backpropagated loss gradient – a sufficient statistic that compresses everything downstream layers contribute to the weight updates affecting layer ℓ . In vector form, $\Delta \mathbf{A}^{(\ell)} \approx -\eta \Phi^{(\ell)} \nabla_{\mathbf{A}^{(\ell)}} \mathcal{L}$: gradient descent in weight space induces *kernel descent* within each layer of activity space. It holds for any differentiable sequential feedforward network; Architectures with skip connections require additional treatment, as weights can influence the loss through paths that bypass intermediate layers.

Relation to the neural tangent kernel. The first-order approximation we study should be familiar to those acquainted with the neural tangent kernel (NTK) (Jacot et al., 2018). The kernel $\Phi^{(\ell)}$ is a layer-wise analogue of the empirical neural tangent kernel (NTK) for finite-width networks (Lee et al., 2020).

Self-terms drive each neuron toward its loss gradient; cross-terms interfere. The diagonal element $\Phi_{ii}^{(\ell)} = \|\nabla_{\mathbf{W}_{\leq \ell}} A_i^{(\ell)}\|^2$ collects contributions from *all* weights that influence neuron i . Because $\Phi_{ii}^{(\ell)} \geq 0$, the self-term update is aligned with $-\partial \mathcal{L} / \partial A_i^{(\ell)}$: each such neuron’s activity moves down its own loss gradient, with an effective learning rate $\eta \Phi_{ii}^{(\ell)}$ that grows with the depth and width of its input pathways.

The off-diagonal elements $\Phi_{ik}^{(\ell)} = \nabla_{\mathbf{W}_{\leq \ell}} A_i^{(\ell)} \cdot \nabla_{\mathbf{W}_{\leq \ell}} A_k^{(\ell)}$ for $k \neq i$ couple neurons *within the same layer*. Two neurons whose activities depend on overlapping upstream weights can

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have large $|\Phi_{ik}^{(\ell)}|$, whereas neurons supported by effectively orthogonal parameter directions have $\Phi_{ik}^{(\ell)} \approx 0$. These cross-terms need not align with $-\partial\mathcal{L}/\partial A_i^{(\ell)}$.

Cross-terms are small when different neurons’ activities depend on nearly orthogonal directions in upstream weight space. Width can increase each neuron’s private incoming parameters, but shared upstream weights mean that activity gradients need not become independent. Greater depth increases the number of shared upstream weights. Learning could have heterogenous effects, and may decorrelate the pathways leading to each neuron.

Activity descent for diagonal-dominant kernels. In the case where $\Phi^{(\ell)}$ is diagonal-dominant, the kernel identity (Equation (4)) yields a simple approximation:

$$\Delta A_i^{(\ell)} \approx -\eta \Phi_{ii}^{(\ell)} \frac{\partial\mathcal{L}}{\partial A_i^{(\ell)}}. \quad (6)$$

Just as weight descent says *strengthen each synapse in proportion to how much it helps*, activity descent says *increase each neuron’s firing in proportion to how much that increase would help* (and decrease it when it hurts). Equation (6) therefore identifies a useful approximation regime: when $\Phi^{(\ell)}$ is sufficiently diagonal-dominant, each neuron’s activity moves in the direction that locally reduces the loss, up to a neuron-specific scaling factor $\Phi_{ii}^{(\ell)}$.

Experiments. We test Equations (4) and (6) in ReLU MLPs trained by online SGD (see full details in Appendix A). We can measure whether Equation (6) holds by correlating the observed activity change $\Delta A_i^{(\ell)}$ with the negative loss gradient $-\partial\mathcal{L}/\partial A_i^{(\ell)}$ across neurons. We report both this raw-gradient correlation and the correlation after including the neuron-specific gain $\Phi_{ii}^{(\ell)}$, pooling across all active hidden units and the two output units. The raw correlation is reduced by off-diagonal interference, by heterogeneity in the diagonal gains, and by higher-order Taylor error; the diagonally scaled correlation factors out gain heterogeneity and therefore more directly tests the diagonal-kernel approximation.

Our results show that in ANNs of moderate width and depth, the diagonal approximation is quite good: activity changes and the raw gradient are correlated, and this correlation is further improved by the diagonal scaling. Neither correlation changes monotonically with width in the regime tested, suggesting that widening alone does not eliminate structured interactions between neurons. Both correlations decrease with depth, consistent with neurons in deeper networks sharing more upstream pathways and therefore experiencing stronger cross-neuron interference.

A testable prediction for neuroscience. If a biological circuit performs a gradient-like synaptic update in a diagonal-dominant regime, Equation (6) predicts that each active neuron’s change is aligned with its negative loss gradient. [Richards and Kording \(2023\)](#) proposed a concrete protocol using existing in vivo methods that could test this prediction. Consider a task with a loss or criterion $\mathcal{L}$ which is based on the animal’s measured behavior b . Both neural activity and behavior can be recorded across trials as the animal learns the task. They note that $\partial\mathcal{L}/\partial A_i^{(\ell)}$ can be decomposed into the product $\partial\mathcal{L}/\partial b \cdot \partial b/\partial A_i^{(\ell)}$, where b is some measured behavior. The first factor $\partial\mathcal{L}/\partial b$ is readily available from the task design, and the second factor $\partial b/\partial A_i^{(\ell)}$ can be estimated by stimulating neuron i and measuring the effect on behavior b (this is already done in many experiments via optogenetics).

A key caveat is that activity descent is *necessary but not sufficient* for weight descent. Observing that activities follow their loss gradient constrains the projection of $\Delta\mathbf{W}$ onto

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the row space of the activity Jacobian $\partial \mathbf{A} / \partial \mathbf{W}$, but says nothing about the null space of weight changes that leave activities unchanged on the current input.

Our work constitutes a theoretical and empirical demonstration that gradient descent in weight-space leads to approximate activity descent in artificial neural networks. Our findings suggest that the same relationship can be tested in biological neural networks, and that doing so could provide evidence for or against gradient-like learning rules in the brain.

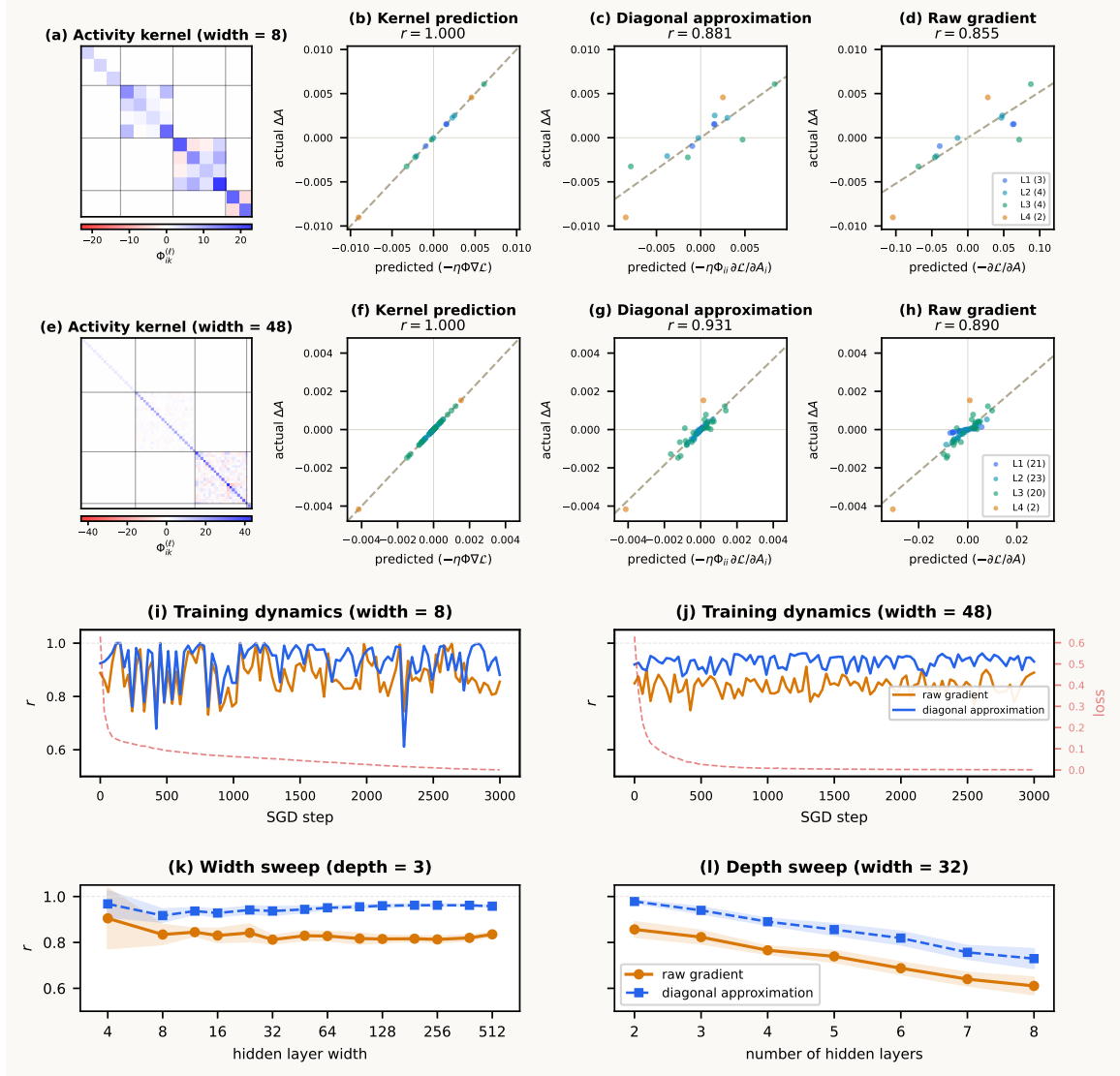


Figure 1: **Kernel descent and activity-gradient alignment.** (a–h) Full-kernel, diagonal, and raw-gradient predictions at widths 8 and 48. (i–j) Raw-gradient and diagonal-approximation correlations through training. (k–l) Median correlations across width and depth; bands show ± 1 s.d. across ten matched replicates.

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Appendix A. Experimental details

All networks are trained by online SGD ($\eta = 0.005$) with squared loss. The training set contains 40 noisy 4×4 horizontal- or vertical-bar images generated from binary templates with independent Gaussian pixel noise ($\sigma = 0.15$; 20 examples per class; input dimension 16; linear output dimension 2). Weights use He initialisation and biases are initialised to zero. The representative width-8 and width-48 networks have three hidden layers and are trained for 3000 SGD steps. The width sweep ranges from 4 to 512 at fixed depth 3, and the depth sweep ranges from 2 to 8 at fixed width 32. Each sweep uses ten matched random replicates trained for 2000 SGD steps; within a replicate, every architecture receives the same generated dataset and sequence of sampled examples. Pearson correlations pool all active hidden and output units; inactive hidden ReLU units are excluded. In the sweep panels, each line is the across-replicate mean of the within-training median and bands show ± 1 s.d. across replicates.

Appendix B. Annotated layer-2 decomposition

To make the diagonal/off-diagonal split concrete, consider a standard layer-2 block (omitting biases for simplicity):

$$z_i^{(2)} = \sum_m W_{im}^{(2)} A_m^{(1)}, \quad A_i^{(2)} = \sigma_2(z_i^{(2)}).$$

The first-order activity update is

$$\Delta A_i^{(2)} \approx -\eta \sum_j \Phi_{ij}^{(2)} \frac{\partial \mathcal{L}}{\partial A_j^{(2)}}.$$

The point of this appendix is to show explicitly where the self term $\Phi_{ii}^{(2)}$ and the cross terms $\Phi_{ij}^{(2)}$ for $j \neq i$ come from.

Own-layer weights contribute only to the diagonal. For a weight $W_{pm}^{(2)}$ feeding directly into layer 2,

$$\frac{\partial A_i^{(2)}}{\partial W_{pm}^{(2)}} = \delta_{ip} \sigma_2'(z_i^{(2)}) A_m^{(1)}.$$

Therefore the layer-2 contribution to the kernel is

$$\sum_{p,m} \frac{\partial A_i^{(2)}}{\partial W_{pm}^{(2)}} \frac{\partial A_j^{(2)}}{\partial W_{pm}^{(2)}} = \delta_{ij} [\sigma_2'(z_i^{(2)})]^2 \|\mathbf{A}^{(1)}\|^2 = \delta_{ij} D_{2,ii}.$$

This term is diagonal because two different neurons in the same layer do not share their own incoming weights.

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Earlier-layer weights are propagated forward by the Jacobian. Now take a weight W_α with $\alpha \in \mathcal{P}_1$. By the chain rule,

$$\frac{\partial A_i^{(2)}}{\partial W_\alpha} = \sum_m \frac{\partial A_i^{(2)}}{\partial A_m^{(1)}} \frac{\partial A_m^{(1)}}{\partial W_\alpha} = \sum_m J_{2,im} \frac{\partial A_m^{(1)}}{\partial W_\alpha}, \quad J_{2,im} = \frac{\partial A_i^{(2)}}{\partial A_m^{(1)}}.$$

Summing over all layer-1 weights gives

$$\sum_{\alpha \in \mathcal{P}_1} \frac{\partial A_i^{(2)}}{\partial W_\alpha} \frac{\partial A_j^{(2)}}{\partial W_\alpha} = \sum_{m,n} J_{2,im} \Phi_{mn}^{(1)} J_{2,jn}.$$

Because layer 1 has no earlier shared weights, $\Phi^{(1)} = D_1$ is diagonal, so this reduces to

$$\sum_{m,n} J_{2,im} \Phi_{mn}^{(1)} J_{2,jn} = \sum_m J_{2,im} D_{1,mm} J_{2,jm}.$$

Thus the full layer-2 kernel is

$$\Phi_{ij}^{(2)} = \delta_{ij} D_{2,ii} + \sum_m J_{2,im} D_{1,mm} J_{2,jm}.$$

If one writes the Jacobian explicitly,

$$J_{2,im} = \sigma'_2(z_i^{(2)}) W_{im}^{(2)},$$

so

$$\Phi_{ij}^{(2)} = \delta_{ij} [\sigma'_2(z_i^{(2)})]^2 \|\mathbf{A}^{(1)}\|^2 + \sigma'_2(z_i^{(2)}) \sigma'_2(z_j^{(2)}) \sum_m W_{im}^{(2)} D_{1,mm} W_{jm}^{(2)}.$$

Self and cross terms in the activity update. Substituting the layer-2 kernel into the first-order update and separating the $j = i$ and $j \neq i$ pieces gives

$$\Delta A_i^{(2)} \approx -\eta \Phi_{ii}^{(2)} \frac{\partial \mathcal{L}}{\partial A_i^{(2)}} - \eta \sum_{j \neq i} \Phi_{ij}^{(2)} \frac{\partial \mathcal{L}}{\partial A_j^{(2)}}.$$

Using the expression above,

$$\Phi_{ii}^{(2)} = D_{2,ii} + \sum_m J_{2,im}^2 D_{1,mm},$$

while for $j \neq i$,

$$\Phi_{ij}^{(2)} = \sum_m J_{2,im} D_{1,mm} J_{2,jm}.$$

Hence

$$\Delta A_i^{(2)} \approx -\eta \left(D_{2,ii} + \sum_m J_{2,im}^2 D_{1,mm} \right) \frac{\partial \mathcal{L}}{\partial A_i^{(2)}} - \eta \sum_{j \neq i} \left(\sum_m J_{2,im} D_{1,mm} J_{2,jm} \right) \frac{\partial \mathcal{L}}{\partial A_j^{(2)}}.$$

The first bracket is the self term: it is always aligned with $-\partial \mathcal{L} / \partial A_i^{(2)}$. The second sum is the interference term: it mixes neuron i with the gradients of other neurons in the same layer through shared earlier-layer parameters.

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Relation to the temporary x_0/x_1 notation. If one writes $x_0 = \mathbf{A}^{(0)}$ and $x_1 = \mathbf{A}^{(1)}$, then

$$D_{1,mm} = [\sigma'_1(z_m^{(1)})]^2 \|x_0\|^2, \quad D_{2,ii} = [\sigma'_2(z_i^{(2)})]^2 \|x_1\|^2.$$

So the appearance of both x_0 and x_1 in scratch work is meaningful, not a typo: they are simply the activities entering layers 1 and 2. In the main text, $\mathbf{A}^{(0)}$ and $\mathbf{A}^{(1)}$ are cleaner and more consistent with the rest of the notation.

Appendix C. General recursion for deeper networks and nonlinearities

For an arbitrary depth- L feedforward network with pointwise nonlinearities,

$$z_i^{(\ell)} = \sum_m W_{im}^{(\ell)} A_m^{(\ell-1)}, \quad A_i^{(\ell)} = \sigma_\ell(z_i^{(\ell)}), \quad \ell = 1, \dots, L,$$

again omitting biases for simplicity. The first-order kernel relation from the main text still holds:

$$\Delta \mathbf{A}^{(\ell)} \approx -\eta \Phi^{(\ell)} \nabla_{\mathbf{A}^{(\ell)}} \mathcal{L}, \quad \Phi_{ij}^{(\ell)} = \sum_{\alpha \in \mathcal{P}_{\leq \ell}} \frac{\partial A_i^{(\ell)}}{\partial W_\alpha} \frac{\partial A_j^{(\ell)}}{\partial W_\alpha}.$$

To obtain a recursion, split the weight sum into layer ℓ itself and layers $1, \dots, \ell - 1$.

Own-layer contribution. For a weight $W_{pm}^{(\ell)}$ in layer ℓ ,

$$\frac{\partial A_i^{(\ell)}}{\partial W_{pm}^{(\ell)}} = \delta_{ip} \sigma'_\ell(z_i^{(\ell)}) A_m^{(\ell-1)}.$$

Therefore the own-layer contribution is diagonal:

$$\sum_{p,m} \frac{\partial A_i^{(\ell)}}{\partial W_{pm}^{(\ell)}} \frac{\partial A_j^{(\ell)}}{\partial W_{pm}^{(\ell)}} = \delta_{ij} [\sigma'_\ell(z_i^{(\ell)})]^2 \|\mathbf{A}^{(\ell-1)}\|^2 = \delta_{ij} D_{\ell,ii}.$$

Earlier-layer contribution. Define the inter-layer Jacobian

$$J_{\ell,im} = \frac{\partial A_i^{(\ell)}}{\partial A_m^{(\ell-1)}} = \sigma'_\ell(z_i^{(\ell)}) W_{im}^{(\ell)}.$$

Then for any weight W_α with $\alpha \in \mathcal{P}_{\leq \ell-1}$,

$$\frac{\partial A_i^{(\ell)}}{\partial W_\alpha} = \sum_m J_{\ell,im} \frac{\partial A_m^{(\ell-1)}}{\partial W_\alpha}.$$

Summing over earlier-layer weights gives

$$\sum_{\alpha \in \mathcal{P}_{\leq \ell-1}} \frac{\partial A_i^{(\ell)}}{\partial W_\alpha} \frac{\partial A_j^{(\ell)}}{\partial W_\alpha} = \sum_{m,n} J_{\ell,im} \Phi_{mn}^{(\ell-1)} J_{\ell,jn}.$$

Combining both pieces yields the recursion

$$\Phi^{(\ell)} = D_\ell + J_\ell \Phi^{(\ell-1)} J_\ell^T, \quad \Phi^{(1)} = D_1.$$

This is the desired recursion in expanded form.

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Unrolling the recursion. Repeated substitution shows that every earlier layer contributes a positive-semidefinite term transported forward by the intermediate Jacobians:

$$\Phi^{(\ell)} = D_\ell + J_\ell D_{\ell-1} J_\ell^T + J_\ell J_{\ell-1} D_{\ell-2} J_{\ell-1}^T J_\ell^T + \cdots + J_\ell J_{\ell-1} \cdots J_2 D_1 J_2^T \cdots J_{\ell-1}^T J_\ell^T.$$

This makes the structure transparent: D_r captures the direct effect of layer- r weights on layer- r activities, and the surrounding Jacobians propagate that contribution forward to layer ℓ .

Role of nonlinearities. Nothing in the kernel definition or recursion requires linear hidden units. For smooth nonlinearities such as sigmoid, tanh, or GELU, the algebraic derivation above holds with the corresponding σ'_ℓ ; the overall $\Delta \mathbf{A} \approx -\eta \Phi \nabla_{\mathbf{A}} \mathcal{L}$ remains a first-order approximation in η . For piecewise-linear nonlinearities such as ReLU or leaky ReLU, the same formulas hold almost everywhere; at kink points one may use a subgradient. In particular, if a ReLU unit is inactive on the sampled input, then $\sigma'_\ell(z_i^{(\ell)}) = 0$, so its row of J_ℓ and its diagonal entry in D_ℓ vanish.

More general feedforward modules. The recursion above assumes an affine layer followed by a pointwise nonlinearity. The kernel definition

$$\Phi_{ij}^{(\ell)} = \sum_{\alpha \in \mathcal{P}_{\leq \ell}} \frac{\partial A_i^{(\ell)}}{\partial W_\alpha} \frac{\partial A_j^{(\ell)}}{\partial W_\alpha}$$

and the first-order relation $\Delta \mathbf{A}^{(\ell)} \approx -\eta \Phi^{(\ell)} \nabla_{\mathbf{A}^{(\ell)}} \mathcal{L}$ still hold for any differentiable sequential feedforward architecture (i.e. one without skip connections). For other modules the same logic applies: there is an own-module contribution from parameters in layer ℓ and a propagated earlier-layer contribution obtained by sandwiching $\Phi^{(\ell-1)}$ between the appropriate Jacobians. Only the explicit forms of D_ℓ and J_ℓ change.